End-to-End RAG Architecture & Tradeoffs

What three 2024–2025 surveys agree on — and where they diverge

One-Slide Summary

- **RAG patches LLM memory** — retrieves external evidence at inference; no retraining needed
- **Three paradigms describe the evolution:** Naive → Advanced → Modular RAG
- **Four design families describe intent:** Retriever-Centric · Generator-Centric · Hybrid · Robustness
- **The Blinkered Chunk Effect** is the central failure of plain retrieval; GraphRAG and RAPTOR fix it
- **Security and data governance are the unsolved frontiers** — defenses lag far behind attacks

The End-to-End RAG Pipeline

<svg viewBox="0 0 800 180" width="100%"> <rect x="10" y="60" width="130" height="60" rx="8" fill="#1a1815" stroke="#b91c1c" stroke-width="2"/> <text x="75" y="87" text-anchor="middle" fill="#f5f0e6" font-size="15" font-family="Georgia">Source Docs</text> <text x="75" y="107" text-anchor="middle" fill="#d4c59a" font-size="13" font-family="Georgia">chunk + embed</text> <rect x="180" y="60" width="130" height="60" rx="8" fill="#1a1815" stroke="#b91c1c" stroke-width="2"/> <text x="245" y="87" text-anchor="middle" fill="#f5f0e6" font-size="15" font-family="Georgia">Vector Store</text> <text x="245" y="107" text-anchor="middle"

Three Generations of RAG

Generation	Era	Core Idea	Weakness
Naive RAG	2020–22	retrieve → generate, fixed pipeline	context loss, noise sensitivity
Advanced RAG	2022–23	query rewriting, HyDE, reranking, compression	pipeline complexity
Modular RAG	2023–	fully reconfigurable components, agent loops	coordination overhead

Four Architectural Intents (Sharma 2025)

- **Retriever-Centric** — RQ-RAG, RAG-Fusion, SEER: innovate at retrieval; generator is passive
- **Generator-Centric** — SELF-RAG, FiD-Light: innovate at decoding; retriever is fixed
- **Hybrid** — DRAGIN, CRAG, IM-RAG: retriever and generator co-adapt at inference time
- **Robustness-Oriented** — RAAT, Bottleneck Filtering: harden against noisy or poisoned context

The two taxonomies are orthogonal, not contradictory

A Modular RAG system can be Hybrid or Robustness-Oriented in intent.

The Blinkered Chunk Effect

> *"A retrieved chunk is right by similarity — but wrong in context."*

- **Plain chunking strips surrounding document meaning** from the retrieved fragment
- *Heavy casualties:* contracts, specs, co-referential knowledge bases, multi-step procedures
- The LLM sees the correct words but cannot reconstruct what came before or after

Architectural mitigations:

Fix	Mechanism
RAPTOR	Hierarchical summary tree — retrieval at the

What RAG Actually Delivers

- **Multi-hop QA (HotpotQA):** RQ-RAG >*800%* over raw LLM; IM-RAG +5.3 F1
- **Short-form QA (PopQA):** SELF-RAG >270%; Self-CRAG *320%* over raw LLM
- **Hallucination reduction:** FILCO −*64%* hallucinations; SEER *9.25*× context reduction with +13.5 F1
- **Knowledge-graph RAG:** KRAGEN −*20–30%* hallucinations vs. text-only RAG
- **Adaptive retrieval:** DRAGIN/FLARE prune ~*15%* redundant fetches with no accuracy loss

Gains compound: hybrid retrieval + context filtering is the dominant production pattern

retrieval augmented generation · multi hop reasoning

Adversarial Threats — A Growing Attack Surface

- **BadRAG (corpus poisoning):** *98.2% attack success rate* with only *0.04% corpus corruption*
- **TrojanRAG:** embedding-level backdoors invisible to content-layer sanitization
- RAG introduces a runtime dependency on *external, potentially untrusted content* — bypassing model-level RLHF defenses
- Best current defense (RAAT adversarial pretraining): +20–30% F1/EM on noisy inputs
- **No system fully defends against high-precision BadRAG-style triggers today**

Small corpus compromise → large, reliable output control

adversarial rag attacks

Three Fundamental Design Tradeoffs

<svg viewBox="0 0 800 200" width="100%"> <rect x="10" y="30" width="240" height="140" rx="10" fill="#1a1815" stroke="#b91c1c" stroke-width="2"/> <text x="130" y="62" text-anchor="middle" fill="#f7e98c" font-size="16" font-family="Georgia" font-weight="bold">Precision vs. Flexibility</text> <text x="130" y="90" text-anchor="middle" fill="#d4c59a" font-size="14" font-family="Georgia">Tighter retrieval</text> <text x="130" y="110" text-anchor="middle" fill="#d4c59a" font-size="14" font-family="Georgia">→ less generative range</text> <text x="130" y="150" text-anchor="middle" fill="#8a8377" font-size="13" font-

Open Questions & Next Steps

- **Measure business impact** — does RAG lift organisational performance or IT–business alignment? No empirical data yet
- **Harden security defenses** — BadRAG attacks are documented; robust corpus-integrity verification does not yet exist
- **Multi-modal RAG** — image, audio, video retrieval flagged as future work across all three surveys; no coverage today
- **Production benchmarks** — latency, cost, index scalability at real-world scale are gaps in all surveyed literature
- **Evaluation depth** — ARES, RAGAS, RAGTruth need domain-specialised and multi-document retrieval stress tests

§ 10 · TAKEAWAY

The One Thing to Remember

RAG's retrieval layer is both its greatest asset and its most exploitable surface.

Context filtering and hybrid retrieval are the highest-ROI improvements available today. GraphRAG and RAPTOR are the architectural bets worth watching for complex, relational corpora. Security and data governance are where the field has the most ground to cover.

Reference: project overview — full synthesis of all three surveys

Sources: rag comprehensive survey 2506 00054 · rag comprehensive survey 2410 12837 · retrieval augmented generation klesel wittmann 2025